



VII Semester BTECH ENDSEM EXAM

Introduction To Extended Reality Application Development and Digital Twins [IT_4325]

Time Duration: 9.30AM to 12.30PM

Date: 16/11/2025

Max marks: 50 MARKS

Question No	Topic	Marks
1. A	Explain the concept of Extended Reality and summarize how it integrates with Digital Twin technology. Points expected: • Definition of XR: Umbrella term covering VR, AR, MR. (1) • Digital Twin: Virtual replica of physical system updated in real time. (1) • Integration: (1) ○ XR visualizes and interacts with the digital twin in immersive 3D. ○ Real-time sensor data → twin → XR environment. ○ Enables simulation, monitoring, remote diagnostics, predictive maintenance.	3
1. B	Differentiate among the four types of modelling used in XR and illustrate their relevance in rendering. Four modelling types: 1. Geometric modelling – shapes/meshes; foundational for rendering geometry. 2. Kinematic modelling – movement/animation; ensures realistic motion. 3. Physical modelling – physics laws; enables realistic collisions, forces. 4. Behavioural modelling – AI, interactivity; drives dynamic scene responses. 2 marks – Any four correct modelling types (geometric, kinematic, physical, behavioural). 1 mark – Relevance of modelling to rendering/realism. Partial marks: 1.5 marks if only 2–3 modelling types with relevance.	3
1. C	Scenario: You are part of a startup building a remote-training simulator using VR headsets. Analyze the essential hardware components required to achieve realistic immersion and justify your selection. Expected points: • HMD with high resolution, high FOV, high refresh rate. • Powerful GPU/PC or standalone headset with strong SoC. • 6DOF tracking system (inside-out cameras or external lighthouse). • Input devices: controllers, hand-tracking sensors. • Audio: spatial headphones/microphones.	4

	• Optional: haptic gloves, treadmills for locomotion. Justification: Increases presence, reduces latency, enhances precision for training. Mark Breakdown: • 1 mark – HMD characteristics (resolution/FOV/refresh rate). • 1 mark – Tracking system (6DOF, inside-out, lighthouse). • 1 mark – Interaction devices (controllers/hand tracking). • 1 mark – Justification for immersion (latency, accuracy, realism). Partial: • Generic hardware list without justification → max 2 marks.	
2. A	Explain the concept of embodiment and discuss how it affects user presence in the Metaverse. Embodiment: User's sense of owning a virtual body (avatar). Effects on presence: • Increased identity alignment. • More natural interaction; reduced cognitive load. • Enhances social presence with avatars. • Sensorimotor synchrony → stronger immersion. Mark Breakdown: • 1 mark – Definition of embodiment. • 2 marks – Explanation of embodiment's effect on presence (identity, agency, sensorimotor alignment). • 1 mark – Example or clear explanation linked to metaverse context.	4
2. B	Illustrate how distributed VR architectures enable interaction between multiple users. • Networked VR connects multiple clients via shared state sync. • Techniques: replication, update broadcasts, state prediction. • Server or P2P maintains world consistency. • Allows real-time avatar movement, voice, object interaction. Mark Breakdown: • 1 mark – Explanation of distributed VR. • 1 mark – Mechanisms (state sync, replication, networking). • 1 mark – How interaction happens between users (avatars, shared world).	3
2. C	Scenario: You are planning a VR training platform for remote teams. Analyze whether a client-server or peer-to-peer architecture would be more efficient and justify your answer. • Central authority, consistent state, secure, scalable → preferred. P2P: • Lower latency but poor synchronization, security issues. Conclusion: Client-server is more efficient for structured training with many users.	3

	Mark Breakdown: • 1 mark – Features of client-server. • 1 mark – Features of P2P. • 1 mark – Justification of which architecture is more efficient.	
3. A	Explain multimodal interaction in XR and identify advantages of gesture-based interfaces. Multimodal interaction: Combination of inputs such as gesture, voice, gaze, controllers, haptics. Gesture advantages: • Natural and intuitive. • Hands-free interaction. • Better immersion. • Suitable when physical controllers are not feasible. Mark Breakdown: • 1 mark – Definition of multimodal interaction. • 2 marks – Any two advantages of gesture interfaces (intuitive, hands-free, immersion).	3
3. B	Discuss how graphics, sound, and haptic feedback displays enhance user immersion. Graphics: Realistic lighting, textures, physics → visual presence. Sound: Spatial audio → directionality, environment awareness. Haptics: Touch feedback → realism, embodiment, task learning. Mark Breakdown: • 1 mark – Graphics. • 1 mark – Sound. • 1 mark – Haptics.	3
3. C	Scenario: In a VR rehabilitation system, users interact with virtual objects using hand gestures. Analyze how navigation and manipulation interfaces can be designed to maximize comfort and minimize fatigue. Points expected: • Use large, simple gestures to reduce strain. • Limit reach distance; keep interactions within comfort zone. • Use assisted navigation (teleport, joystick). • Provide gravity/physics cues for easier object manipulation. • Adaptive difficulty to reduce fatigue. • Use rest breaks and ergonomic design. Mark Breakdown: • 2 marks – Navigation design (teleport, comfort, reduced strain). • 2 marks – Manipulation design (gesture size, reach zone, ergonomics).	4
4. A	Describe multiple-camera tracking and discuss its application in full-body motion capture. • Multiple cameras observe markers or features from different angles. • Triangulation computes 3D positions. • High accuracy for full-body skeleton reconstruction. • Used in animation, VR avatars, biomechanics.	4

	Mark Breakdown: • 1 mark – What multi-camera tracking is. • 2 marks – How triangulation / 3D reconstruction works. • 1 mark – Application in motion capture / XR avatars.	
4. B	Explain how IMU and lighthouse systems improve spatial tracking accuracy. IMU: Measures acceleration & gyroscope rotation → fast but drifts. Lighthouse: External base stations give precise optical tracking → no drift. Combination: Fuses fast IMU data + precise optical data → high accuracy, low latency. Mark Breakdown: • 1 mark – Role of IMU. • 1 mark – Role of lighthouse. • 1 mark – Fusion benefits (high precision + low latency).	3
4. C	Scenario: In an outdoor AR exploration app, evaluate how GPS-based tracking compares with SLAM for position accuracy GPS: • 3–5 m accuracy; good outdoors; poor for fine AR alignment. SLAM: • Uses camera to map environment → centimeter-level precision. • Works poorly in wide-open textureless spaces. Conclusion: SLAM offers much better positional accuracy for AR; GPS only suitable for coarse location. Mark Breakdown: • 1 mark – GPS characteristics. • 1 mark – SLAM characteristics. • 1 mark – Accuracy-based evaluation/conclusion.	3
5A.	Explain the role of a game engine in designing virtual worlds and summarize its key components. Role: Creates and manages virtual worlds for XR. Components: • Rendering engine • Physics engine • Scene graph • Animation system	3

	• Scripting system • UI, input, audio modules 1 mark – Role of game engine in XR. 2 marks – Any two components (rendering, physics, scripting, audio, animation).	
5B.	Demonstrate how scripting in Unity (C#) and Unreal (Blueprint) enhances interactivity in XR applications. Unity (C#): • Event-driven scripts add behaviors (collisions, interactions, game logic). Unreal (Blueprint): • Node-based visual scripting for interactions, animations, triggers. Common: Faster prototyping, dynamic behaviour, modular XR interactions. Mark Breakdown: • 1.5 marks – Unity C# scripting explanation + example. • 1.5 marks – Blueprint visual scripting explanation + example.	3
5 C	Scenario: You are leading a team developing a mixed-reality educational game. Design a brief approach outlining how physics simulation and animation contribute to an engaging learning experience. Points expected: • Use physics to simulate real-world cause–effect (gravity, forces). • Interactive experiments → students learn by doing. • Animations show processes (molecular motion, historical events). • Blend MR anchors and virtual objects to maintain stability. • Smooth transitions, feedback animations for engagement. Mark Breakdown: • 2 marks – Physics simulation for learning (cause–effect, real-world behaviour). • 2 marks – Animation for clarity, feedback, engagement. Partial marks: • Only physics or only animation → max 2.	4